

LG-25-082

May 2, 2025

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Limerick Generating Station, Unit 2
Renewed Facility Operating License No. NPF-85
NRC Docket No. 50-353

Subject: LER 2025-001-00 HPCI Valve Isolated During a Surveillance Test Due to a Degraded Test Connector

In accordance with the requirements of 10 CFR 50.73(a)(2)(v)(D), Limerick Generating Station hereby submits the enclosed Licensee Event Report.

There are no commitments contained in this letter.

If you have any questions, please contact Jordan Rajan at (610) 718-3400.

Respectfully,



Michael F. Gillin
Vice President – Limerick Generating Station
Constellation Energy Generation, LLC

cc: Administrator Region I, USNRC
USNRC Senior Resident Inspector, Limerick Generating Station



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form)

<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollcts.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Limerick Generating Station, Unit 2

☒ 050☐ 052

2. Docket Number

353

3. Page

1 OF 3

4. Title

HPCI Valve Isolated During a Surveillance Test Due to a Degraded Test Connector

5. Event Date

Month	Day	Year
03	03	2025

6. LER Number

Year	Sequential Number	Revision No.
2025	- 001 -	00

7. Report Date

Month	Day	Year
05	02	2025

8. Other Facilities Involved

Facility Name	☐ 050	Docket Number
Facility Name	☐ 052	Docket Number

9. Operating Mode

1

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(ii)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(ii)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact
Jordan RajanPhone Number (Include area code)
610-718-3400

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	BJ			Yes					

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On March 03, 2025, at 10:44 during planned surveillance testing, the High Pressure Coolant Injection (HPCI) system Turbine Exhaust Line Vacuum Breaker Outboard primary containment isolation valve (PCIV) closed. The valve closed due to a degraded cannon plug connector that caused a short within the Emergency Core Cooling System (ECCS) Contact Test Unit being used for the test. The test equipment was removed, and the valve was re-opened. This event is being reported because it resulted in an unplanned inoperability of a single train safety system that could have prevented fulfillment of a safety function. An eight (8) hour ENS notification was completed on March 03, 2025, at 18:09 (ENS 57582).

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME

Limerick Generating Station, Unit 2



050



052

2. DOCKET NUMBER

353

3. LER NUMBER

YEAR

2025

SEQUENTIAL

NUMBER

001

REV

NO.

00

NARRATIVE**Unit Condition Prior to the Event**

Prior to the event, Unit 2 was in Operational Condition (OPCON) 1 (Power Operation) at approximately 100% power. There were no other structures, systems, or components out of service that contributed to this event.

Description of the Event

On March 03, 2025, a surveillance test was being conducted for Unit 2 High Pressure Coolant Injection (HPCI) System [BJ] to verify that the HPCI logic circuit actuated on a simulated HPCI Steam Supply Pressure- Low signal. At 10:44, the HPCI Turbine Exhaust Line Vacuum Breaker Outboard primary containment isolation valve (PCIV) closed and the HPCI system was declared inoperable.

The test was aborted by removing the ECCS Contact Test Unit (test box), resetting the isolation, and restoring operability of the system. The HPCI System was returned to service on March 03, 2025, at 13:51. An eight (8) hour ENS notification (57582) was completed on March 03, 2025, at 18:09 hours.

Analysis of the Event

The HPCI System was operable during the performance of the instrument testing. The Limerick Technical Specifications allows the system to remain operable for up to eight hours during the instrument testing provided at least one operable channel in the same trip system is monitoring that parameter.

During the instrument testing, when the isolation signal was actuated (Low Steam Line Pressure) the HPCI Turbine Exhaust Line Vacuum Breaker Outboard PCIV closed rendering HPCI inoperable. Immediately following the isolation, the ECCS Contact Test Unit was removed from the HPCI System circuit by unplugging the cannon plug from the panel. Resistance checks were performed of the unit and one electrical short circuit was found. This electrical short circuit resulted in the closure of HPCI Turbine Exhaust Line Vacuum Breaker Outboard PCIV.

Safety Consequence

There was no actual safety consequences associated with this event and the potential safety consequences of this event were minimal. The Reactor Core Isolation (RCIC) System, Automatic Depressurization System (ADS), Residual Heat Removal (RHR) System, and Core Spray (CS) System remained operable during the period that HPCI was inoperable.

This condition resulted in a loss of safety function for the HPCI System and is reportable per 10 CFT 50.73(a)(2)(v)(D).

Cause of the Event

The cannon plug connector was degraded resulting in a short within the ECCS Contact Test Unit.



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1. FACILITY NAME Limerick Generating Station, Unit 2	☒ 050	2. DOCKET NUMBER 353	3. LER NUMBER		
	☐ 052		YEAR 2025	SEQUENTIAL NUMBER 001	REV NO. 00

NARRATIVE

Corrective Actions Completed

The ECCS Contact Test Unit was immediately quarantined and the cannon plug connector was replaced.

Corrective Actions Planned

- Replace cannon plug connectors on all ECCS Contact Test Units.
- Evaluate change in PM strategy/testing on ECCS Contact Test Units.

Previous Similar Occurrences

LER 1-2022-001 Unit 1 HPCI Inoperable Due to Inadvertent Isolation Signal

Component Data:

System: BJ High Pressure Coolant Injection System